HARIPRASATH G

Software Engineer • AI/ML Engineer • Full-Stack Developer • LLM Fine-Tuning & Agentic Systems
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PROFESSIONAL SUMMARY

- AI/ML Engineer and Software Engineer specialising in LLM fine-tuning (QLoRA/PEFT), deep learning, production RAG pipelines, and autonomous multi-agent systems powered by Model Context Protocol (MCP).
- Proven full-stack developer with hands-on industry experience delivering scalable MERN applications, Redis-cached RESTful APIs, system design, and cloud-native backends in fast-paced Agile/Scrum environments.
- Independently architected enterprise-grade AI platforms — on-premise agentic hubs, fine-tuned multi-modal systems, and distributed agent orchestration frameworks deployed on Cloudflare and Kubernetes.
- Passionate about engineering intelligent, privacy-critical, production-ready software systems through strong problem-solving, data structures & algorithms, and continuous learning.

EDUCATION

B.Tech – Artificial Intelligence & Data Science | Arunai Engineering College, Anna University *2022 – 2026*
Tamil Nadu, India

EXPERIENCE

Web Trendz Technologies India Pvt. Ltd. — MERN Stack Developer Intern *Jun 2025 – Aug 2025 (3 Months)*
React.js • Node.js • Express.js • MongoDB • Redis • REST APIs • Tailwind CSS • GitHub Actions • Agile/Scrum

- Spearheaded full-stack development of 3 production-deployed web applications using React.js and Tailwind CSS, improving page load performance by 30% and achieving cross-browser compatibility across all major platforms.
- Architected and implemented 10+ RESTful API endpoints with Node.js and Express.js; integrated Redis caching layer that reduced redundant database hits by 45% and significantly improved API response times under load.
- Applied system design principles to optimise MongoDB collections and indexing strategies, cutting average query response time by 40% and sustaining high throughput under concurrent user loads.
- Established Git branching workflows and CI/CD pipelines via GitHub Actions, reducing manual deployment overhead by 50% and enabling zero-downtime continuous releases to production.
- Collaborated in Agile/Scrum ceremonies including sprint planning, daily standups, and retrospectives; authored comprehensive API documentation and technical specs that improved team onboarding efficiency.

PROJECTS

Hask AI — Fine-Tuned LLM Multi-Modal AI Platform *Aug 2025 – Present*
React • Node.js • MongoDB • Redis • QLoRA • PEFT • Deep Learning • RAG • LangChain • ChromaDB • NLP • OAuth 2.0 • JWT

- Fine-tuned a domain-specific LLM using QLoRA + PEFT deep learning techniques on a curated NLP instruction dataset, achieving domain benchmark improvements while cutting GPU VRAM by ~60% via 4-bit NF4 double quantisation.
- Engineered a production RAG pipeline (LangChain + ChromaDB vector store + BM25 hybrid retrieval) with Redis semantic caching delivering sub-200ms retrieval latency and hallucination-free responses over private document corpora.
- Secured platform with OAuth 2.0 + JWT authentication; integrated fine-tuned diffusion model for multi-modal image generation with real-time WebSocket chat within a scalable, containerised MERN backend.

NeuroOps — On-Premise AI Hub for Secure Workflow Automation *2025 – Present*
Python • LangChain • LlamaIndex • FastAPI • MCP • Docker • Kubernetes • Redis • Cloudflare Zero Trust • System Design

- Architected an enterprise-grade, air-gapped AI platform integrating LLM reasoning, RAG memory, and MCP-powered multi-agent orchestration — zero external API dependency, purpose-built for privacy-critical industries.
- Integrated Model Context Protocol (MCP) servers for standardised tool-use across agents; leveraged Redis for fast agent state caching and session memory; applied distributed system design for scalable workflow execution.
- Deployed Cloudflare Zero Trust tunnels for secure authenticated remote access; implemented RBAC, audit trails, and Docker/Kubernetes orchestration for compliance-ready, production-grade execution.

AgentForge — Autonomous Multi-Agent Orchestration Framework *2025*
Python • LangChain • ReAct • FastAPI • MCP • ChromaDB • Redis • Cloudflare Workers • API Development

- Built a modular multi-agent framework using ReAct reasoning and MCP-connected tool-use (web search, code execution, API integrations) with autonomous task decomposition and self-correcting feedback cycles.
- Implemented persistent agent memory with episodic + semantic layers (ChromaDB) and Redis-backed working memory, enabling agents to retain context across sessions and improve decision quality over successive runs.
- Deployed orchestration API via FastAPI with real-time agent streaming, inter-agent communication bus, and a tracing dashboard on Cloudflare Workers for globally distributed, low-latency agentic execution.

TECHNICAL SKILLS

- **AI / ML:** Machine Learning, Deep Learning, Neural Networks, NLP, LLM Fine-Tuning (QLoRA, PEFT, LoRA), RAG, LangChain, LlamaIndex, MCP (Model Context Protocol), Autonomous Agents, ReAct, Prompt Engineering, Hugging Face, ChromaDB, FAISS, Vector Databases
- **Backend:** Node.js, Express.js, FastAPI, RESTful API Development, Microservices, WebSockets, OAuth 2.0, JWT, MCP Servers, System Design
- **Frontend:** React.js, React Native (Expo), TypeScript, JavaScript, Tailwind CSS, HTML5, CSS3
- **Databases:** MongoDB, PostgreSQL, Redis, ChromaDB, FAISS
- **DevOps / Cloud:** Docker, Kubernetes, CI/CD Pipelines, GitHub Actions, GCP, Cloudflare Workers, Cloudflare Zero Trust
- **Languages:** Python, JavaScript, TypeScript, Java 8+, SQL, Data Structures & Algorithms (DSA)
- **Methodologies:** Agile, Scrum, Full-Stack Development, Software Engineering, Problem Solving, Code Review, Technical Documentation

CERTIFICATIONS & ACHIEVEMENTS

- MERN Stack Developer Intern Certificate — Web Trendz Technologies India Pvt. Ltd. | IoT Internship Certificate — Praya Labs
- National Level Hackathon (24 hrs.) 2nd Prize | CSI Award — Innovation Project Award